Parag Sarvoday Sahu

Research Internship in 3D Computer Vision & Neural Rendering
MSc&T Visual & Creative AI, École Polytechnique | Available 22 March to 31 August 2027

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Master's student at École Polytechnique working on 3D vision, inverse rendering and Gaussian splatting, with a Pacific Graphics 2025 paper, a first-author WACV 2027 submission, and 2nd place worldwide in the SIGGRAPH Asia 2025 3DGS Challenge. Core skills: differentiable rendering and PyTorch research code. English (fluent), Hindi (native), French (A1, learning). Seeking a research internship from 22 March to 31 August 2027.

Education

École Polytechnique (IP Paris), Palaiseau, France — Sep 2026 – Present
MSc&T in Visual and Creative Artificial Intelligence — French Government Scholar (Charpak)
Courses: Digital Representation & Analysis of Shapes, Computer Animation, Foundations of ML

Indian Institute of Technology (IIT) Gandhinagar, India — 2022 – 2026
B.Tech in Electrical Engineering, Minor in Computer Science — CPI **9.04**/10, Dean's List
Courses: Computer Vision, Machine Learning, Matrix Methods for ML, Probability & Statistics

Publications

TensoIS: Feed-Forward Tensorial Inverse Subsurface Scattering for Perlin Distributed Heterogeneous Media
A. Tiwari, S. Bhardwaj, Y. Bachwana, **P. S. Sahu**, S. Raman — *Pacific Graphics 2025 (Computer Graphics Forum)*

Manuscript on Multi-View Photometric Stereo with 2D Gaussian Splatting
P. S. Sahu, A. Tiwari, V. Shah, S. Raman — *First author, under review at WACV 2027*

Research Experience

IndiaAI Research Fellow (National Merit Fellowship) — Jan 2026 - Present
IIT Gandhinagar | Inverse Rendering with Gaussian Primitives — Advisor: Prof. Shanmuganathan Raman
- Formulated a novel 2D Gaussian Splatting approach to uncalibrated Multi-View Photometric Stereo (MVPS), jointly recovering geometry and materials without known lighting directions or intensities.
- Achieved high-fidelity relighting and competitive normal estimation (MAE) while uniquely supporting view-unaligned captures across varying illuminations.
- Reduced the required light captures by **10**× and accelerated optimization from 12 hours to under 1 hour per object, culminating in a first-author WACV 2027 submission.

SIGGRAPH Asia 2025 3DGS Challenge, Hong Kong — Oct 2025 – Nov 2025
Real-time 3D Gaussian Splatting for novel view synthesis — paragsarvoday/DeadlineDinosaur
- Ranked **2nd of about 50 teams worldwide** with **28.04 dB** mean PSNR over 30 test scenes, using a coarse-to-fine progressive-resolution schedule that trains each scene in under 60 seconds.
- Secured a **$2,000 USD** prize and presented our methodology at the SIGGRAPH Asia 3DGS Workshop, focusing on our choice of backbone and careful experimentation with hyperparameters.

Research Intern, Jodhpur, India — Jul 2025 – Dec 2025
3DVisLab, IIT Jodhpur — Advisor: Prof. Avinash Sharma
- Explored learning-based 3D reflective-symmetry detection as a basis for symmetry-aware mesh compression.
- Applied mesh processing and deep learning to high-fidelity 3D human head reconstruction.

Vision Transformer Inference Engine from Scratch (personal project) — Mar 2026 – Apr 2026
Pure C++ / Eigen, deployed to the browser with WebAssembly — paragsarvoday/ViT-Inference
- Implemented ViT inference without a deep-learning framework, shipping an on-device web demo.

Languages

English: Fluent (Working language) **Hindi:** Native **French:** A1 (CEFR, currently studying)

IT Skills

Programming: Python, C/C++, WebAssembly, MATLAB
Libraries & Tools: PyTorch, Eigen, OpenCV, Git/GitHub, LaTeX

Honors and Awards

- **IndiaAI Fellowship:** National merit grant supporting independent research in inverse rendering. 2026
- **Charpak Master's Scholarship:** Selective French government award for graduate studies (1 of 19 nationwide). 2026
- **Bipin & Rekha Shah & Prof. DV Pai Scholarships:** Awarded for academic excellence at IIT. 2023, 2024
- **JEE Advanced (Top 1%):** Ranked in the top 1 percentile of over 1 million candidates nationwide. 2022

Interests

Badminton **Reading:** paragsarvoday.github.io/books